Definition of AK and risk of progression to iSCC

Consensus statements

- Actinic keratosis is a condition encompassing squamous cell carcinoma (SCC) ab initio, including SCC in situ (SCCIS), at different stages of carcinogenesis.
- Lesion-wise, the risk of progression to invasive SCC (iSCC) is relatively low; however, the patient-wise lifetime risk is appreciable.

Actinic keratosis (AK) is a skin disease that typically develops in sun-exposed areas. Lesions can present as macules, papules, or plaques, which may be pigmented or non-pigmented. The affected skin often feels rough and dry. Clinical appearances include flat, scaly lesions or exophytic forms—such as wart-like or hyperkeratotic horn-like lesions. Most patients exhibit one or more areas of photodamaged skin with multiple AKs. While most AKs are asymptomatic, some individuals report pruritus, burning, or a splinter-like sensation.

Several clinical and histological variants of AK have been described, including pigmented, hypertrophic, atrophic, lichenoid, proliferative, acantholytic, and Bowenoid subtypes. These variants can complicate diagnosis due to overlapping features with other dermatoses. Although various diagnostic criteria and classification systems exist for AK subtypes, a comprehensive evaluation of differential diagnosis schemes was beyond the scope of this review.

Given the morphological, genetic, and clinical characteristics of AK, there has been ongoing debate about whether it should be classified as an intraepithelial dysplasia of varying grade or as a neoplastic condition involving atypical keratinocyte proliferation. Strong evidence supports a histopathological continuum between AK and SCCIS: in patients with AK, keratinocytic atypia may range from focal involvement of basal or supra-basal layers to full-thickness epidermal atypia—consistent with SCCIS. Furthermore, up to 90% of invasive SCCs arise in association with or directly from AK lesions.

These morphological parallels reflect shared pathogenic mechanisms. Both AK and iSCC involve a multistep process driven by cumulative UV exposure, p53 gene mutations, impaired DNA repair, immune dysregulation, and stromal changes that promote tumor progression. The karyotypic profile of AK resembles that of iSCC, though AK generally exhibits a lower mutational burden and genomic complexity, consistent with an earlier stage in tumorigenesis. Additionally, the biological activity of complement factors H and I—natural inhibitors of immunity implicated in iSCC invasion—shows no significant difference between SCCIS and AK, further underscoring their biological similarity.

Despite this evidence, some pathologists and clinicians still classify AK as a precancerous condition, citing the potential for spontaneous regression and the low progression risk per individual lesion. However, the patient-level risk of progression to iSCC can reach up to 10% over 10 years in individuals with an average of 7–8 lesions. Importantly, studies on the natural history of AK demonstrate a graded increase in progression risk with higher lesion counts, without evidence of a threshold effect.

Diagnosis

Summary of recommendations

- The diagnosis of AK should be based on clinical appearance and dermoscopic assessment.
- Biopsy is recommended when differential diagnosis with basal cell carcinoma (BCC), Bowen's disease, iSCC, amelanotic melanoma, lentigo maligna melanoma (LMM), or superficial seborrheic keratosis cannot be established by clinical and dermoscopic evaluation alone. Alarm signs should guide decisions toward invasive diagnostic procedures.

The diagnosis of AK is traditionally made through clinical examination. However, clinical assessment alone is often inaccurate. Dermoscopy, although operator-dependent, improves diagnostic accuracy and can effectively differentiate both pigmented and non-pigmented skin lesions, including AK (sensitivity: 99%; specificity: 95%). Among non-invasive tools, reflectance confocal microscopy (RCM) enables high-resolution, cellular-level visualization of the epidermis and superficial dermis and holds potential to complement or even substitute histopathological examination in selected cases.